

ITC C103
WW1 - M2U1_U2 Prelims Assignment 4

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BSIT - 102I

Questions:

1. Explain the relationship between binary notation and the physical components of a computer, such as transistors. Why is binary notation the fundamental language of computers, and how does this concept apply to the digital representation of all types of data, including text, images, and sound?

Answer: Transistors are on/off switches, and binary's 1s and 0s map directly to those two states, making it the natural language of computer hardware. All data types (text, images, sound) are ultimately converted into these binary patterns for the hardware to process.

2. Our lesson describes hexadecimal notation as a more efficient way to represent large values than binary. Describe a real-world scenario in which hexadecimal notation is used, and explain why it is more practical for humans to use in that context.

Answer: Web color codes use hex because one hex digit represents 4 binary digits, making long binary strings much shorter and easier for humans to read and type.

3. Our lesson discusses several forms of intellectual property, including copyright, trademark, and patent. Choose one of these forms of protection and explain a time when you have encountered it in your daily life, such as in a software application, a logo, or a product design.

Answer: The Nike "swoosh" logo is a trademark, it legally prevents other companies from using a similar logo, protecting brand identity and helping consumers trust the product's source.

4. Our lesson states that data is used to drive strategic business decisions in areas like sales and marketing. Provide a specific example of how a company could use data analytics to improve its marketing campaign for a product you are familiar with.

Answer: A fitness apparel company could analyze browsing vs. purchase data to see where customers drop off, then send targeted promotions instead of generic ads.

5. Explain the difference between a bit, a byte, a kilobyte, and a gigabyte, and provide a real-world example of a file that would be measured in each of these units. Why is it important to understand these units when dealing with a computer's storage capacity?

Answer: A bit is a single 0 or 1; a byte (8 bits) can represent one character; a kilobyte (1,000 bytes) fits a short document; a gigabyte (1 billion bytes) fits an HD movie. Knowing these helps you judge storage needs and avoid running out of space.

6. Our lesson describes various types of software licenses, including proprietary, freeware, shareware, and open source. Choose two of these license types and explain how a developer's choice of license could impact both their business model and a user's experience with the software.

Answer: Proprietary software (Like MS Office) is company-controlled and paid, giving developers revenue but limiting user flexibility. Open source (Like Linux) is free and modifiable, benefiting users, but developers must find other ways (support, hosting) to make money.

7. Explain the importance of a proper uninstallation process for an application. What are the potential consequences of simply deleting an application's files from the Program files folder and how does this relate to the concept of "orphaned" files and registry entries?

Answer: Deleting an app's folder alone leaves behind registry entries and files in other system locations ("orphaned" data), which can clutter the system, slow performance, and cause future conflicts, a proper uninstaller removes these too.

8. Our lesson discusses several best practices for installing software, such as checking for OS and hardware compatibility. Describe a hypothetical situation where a user ignores these practices, and explain the specific problems they might encounter as a result.

Answer: If a user installs software without checking OS/hardware compatibility, it may crash, run poorly, or fail to launch, potentially causing data loss and requiring a reinstall or hardware upgrade.

9. Our lesson categorizes software into different types, such as productivity, collaboration, and business software. Choose one specific application (e.g., Google Docs, Microsoft Teams) and describe how it can be used for both productivity and collaboration, providing examples of its features in each context.

Answer: Individually, it's used for writing/formatting documents; collaboratively, multiple users can edit in real time, comment, and suggest changes, making it useful for both solo and team work.

10. What is the purpose of an End-User License Agreement (EULA)? Describe a scenario where a user might encounter a EULA, and explain why it is a legally binding contract that defines the terms of use for the software.

Answer: A EULA is a legal agreement outlining how software can be used; clicking "I Agree" during installation makes it binding, protecting the company's rights while informing users of usage terms.